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Computer Science II

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Final Project Proposal

Project- For our project, we chose to create the game Solitaire using Javafx. The game solitaire has many different forms/versions that are played throughout the world. The one we chose to use was the version known as Klondike. Klondike is the most popular and most simplistic version of solitaire. Klondike is a one-player card game where the player's objective is to arrange the cards into 4 different piles. Each pile begins with an ace and is layered on top of one another in ascending order within the same suit. Once you have achieved 4 different piles from ace to king, the game is then considered to be finished and the user will be granted a score.

Problems to be Solved:

- Creating a card
- Creating a game board
- Storing each cards value
- Storing each cards suit

- Stack the cards on top of the other when it's allowed
- Good looking UI
- Drawing unique random cards from the deck without overlapping
- Creating a screen to show when the user has finished the game.

Topics:

- Object Orientated Programming- In our project, we have multiple classes with different methods which all play a role in creating the solitaire game built into the program. Each class plays a major role in the system. The card class creates each card itself, the card color class sets each card to its own color of either red or black, and the suit color sets each card its own suit from the four available options. In the game class, many card objects have to be created in order to play the game which calls many different instance objects from the card class. Everytime, one of these card objects is created they call on the constructor created within the card class. Throughout the game class there are many reference variables to the other classes within the solitaire project.
- Wrapper classes/ArrayLists- We implemented wrapper classes in the game class when we had to create ArrayLists and maps. We created ArrayLists to make our game board where we would put the cards during the games. Each

card needed its own unique space on the board which were created through the help of ArrayLists. In each ArrayList, we had to use wrappers to construct each and to store it as a variable in it. We also created maps/hashsets to hold our images for each card we wanted to create to better develop the user interface. In the map, we used the wrapper class of String and Image. We also used stacks for the selected cards and hands. In these stack objects, we also implemented object-orientated programming when we used Card objects as our wrapper class for the stacks.

- **Class Constants-** We used class constants in the card and game classes. We used the final keyword to initialize the suit and values of the playing cards within the card class. This made the suit and value of the card construct its final version, meaning it could not be changed later on within the program. In the game class, we used constants to create the lists, stacks, and maps which helped create the user interface of the gameboard and cards.
- **Iterators-** We used iterators to help in looping through our array lists and our hash sets to help create our game boards for the program to execute on.
- **Event-driven Programming-** Throughout our solitaire program we used event-driven programming when the user interacted with each of the cards. When they took a card from a pile the next card would flip over revealing itself to the user in order for him to have a chance to use it. When the user

wanted to move cards to a different pile the program would only allow him to place the cards where it would follow the rules of the game.

Set of Tasks:

We came together and brainstormed ideas for the games we'd like to program. The list of ideas we had were hangman, a dna sequence game, solitaire, and blackjack. We ended up going with solitaire because we both knew a good bit about the game already. After choosing solitaire we divided up the project parts among ourselves. I chose to make or presentation and proposal and she got to do the programming. We didn't plan the greatest to be honest because we waited until last week to discuss. Although we waited so late, we properly discussed our goals and how we wanted to execute our project of solitaire.

Members of the Team:

Trevor- I chose to work on our powerpoint presentation that we will be presenting in class for the final. I also chose to write our project report explaining everything about creating this project and why we did so.

Yasmin- She got to write the actual javafx code for our solitaire project.

Solitaire Program Example:

